Problem Set – Pass by Reference Functions. Create an IPO for each problem within this document. Save the document and upload to Blackboard for grading. Then create a C++ program for each problem. Upload the .CPP files to Blackboard for grading.

1.Allow the user to enter a quantity and price, use ctl+z to stop. Use one function to compute the total (quantity times price), tax (7% of total) and total order (total plus tax). The function should be passed by the quantity and price by value and total, tax and total order by reference. Display total, tax and total order in main. Sum and display total of all orders and tax for all orders and display after the loop (all data is processed).

Input	Process	Output
Quantity	Const double tax_rate = 0.07;	Total:

price	Void calcOrder (int quantity, double price, double &tax, double &totalOrder)
	
	{
	total = quantity * price;
	Tax = total * tax rate
	TotalOrder = total + tax_rate
	
	}
		Tax:

Total Tax Total order	While (cin >> quantity >> price)	Total Order:
Sum total		
Sum tax		

2.Enter the weight of a package and zip code. Use ctl+z to stop. Use a single function to do the computations specified next. Pass these weight and zip code by value, and pass postage, area charge and weight charge by reference. Compute postage to be sum of weight charge and area charge. Use tables below to find the charges. Compute weight charge to be weight x weight charge per ounce. Find the area charge in the table based on zip code. Then compute postage to be area charge plus weight charge. The function should return the weight charge, area charge and postage. Display area charge, weight charge and postage. Count and display the number of entries made.

Area Table – Used to determine the area charge

Area	Area Charge
60171	$2.00
60172	$2.50
60635	$3.00
All others	$5.00

Weight Table – used to determine the weight charge

Weight	Weight Charge per Ounce
>100	0.02
>50	0.03
All other	0.05

Input	Process	Output

weight	Void calcPostage (inputs)
	{
	If/ else statements for area table and weight table
	
	}
		Area Charge:

Zipcode	While (cin >> weight >> zipcode)	Postage:
Postage Area charge Weight charge		Weight charge:

3.Enter the student's last name, credit hours and financial aid, use ctl+z to stop. Pass credit hours and financial aid to a function by value. Pass tuition and tuition owed by reference. Compute tuition to be credit hours times $250. Compute tuition owed to be tuition minus the financial aid. Display student's last name, tuition and tuition owed. Sum and display total tuition owed by all students, count of number of entries and average amount owed by students.

Input	Process	Output
Last name	Const double rate per credit = 250.00	Student last name:
Credit hours Financial aid	Void(Inputs/ & for pass by reference values)	Tuition: Tuititon Owed:
Tuition tuition owed	While (cin >> last name >> credit hours >> financial aide)	Number of entries: Average amount:
Total owed		

4.Enter a number of widgets, use ctl+z to stop. Pass the number to a function by value, use ctl+z to stop. Use a single function to determine the cost per widget using the cost table below. Then compute extended price (number of widgets x cost per widget) and 7% sales tax. Finally compute total order to be extended price plus sales tax. Pass

cost per widget, extended price, sales tax and total order by reference. For each line, display number of widgets, cost per widget, extended price, sales tax and total order. Sum all total orders and display when there is no more data to process.

Cost Table

Number of Widgets	Cost Per Widget
10000 and up	4.00
5,000 and up	5.00
All other amounts	10.00

Input	Process	Output
Number of widgets		
Extended price	void calcOrder(int numWidgets, double &costPerWidget, double &extendedPrice, double &salesTax, double &totalOrder)	Widgets: Widget Cost:

Extended price	while (cin >> numWidgets)
	
	{
	calcOrder(numWidgets, costPerWidget, extendedPrice, salesTax, totalOrder);
	}
		Extended Price:
		Sales Tax:

Sales tax		Order total:
Grand total		

5.Enter the amount of investment, the 5-year interest rate and 10 year interest rate, use ctl+z to stop. Pass the amount and interest rates to a function by value. Pass variables representing five year amount and ten year amount to the same function by reference. Compute the five year amount and ten year amount using the formula below. Display the amount of the investment, the five-year amount and the ten year amount.

Five year amount = amount of the investment x (1 + 5 year rate) raised to 5th power.

Ten year amount = amount of the investment x (1+10 year rate) raised to 10th power.

Note: Enter the 5-year rate and 10 year rate in decimal form, i,e 5% is entered as 0.05.

Also Note: you need to use the pow built in function. Recall the pow function syntax:

pow (base, exponent)

In this case, base is 1 + 5 year rate and exponent is 5.

Another line will be: base is 1 + 10 year rate and exponent is 10.

Use #include<math.h> for the pow function.

Input	Process	Output
Amount		
Rate5	void computeAmounts(double amount, double rate5, double rate10, double &fiveYearAmt, double &tenYearAmt)	Amount Invested:
Rate10	while (cin >> amount >> rate5 >> rate10)	5-Year Investment:
Five year amount		10- year Investment:
10 year amount		